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# Dynamic Characteristics and Structural Response of a Supercavitating Underwater Vehicle: An Experimental and Numerical Study

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**Abstract:** Supercavitating underwater vehicles experience unique stress states and dynamic characteristics compared to conventional underwater vehicles. This study investigates the structural behavior of a supercavitating vehicle under time-varying forces, focusing on the interaction between the nose cavitator and water, and the effects of tail-slapping. A supercavitating flow shell structure model experiment was conducted using a nose-mounted cable towing system, demonstrating significant drag reduction with increasing speed and a corresponding increase in tail-slap vibration. Numerical calculations revealed that the cylindrical shell section experiences the highest stress during tail-slapping, potentially leading to structural failure. High acceleration values during tail-slapping indicate extreme overload conditions. Structural reinforcement of the cylindrical shell is recommended. FEM analysis suggests that all parts of the supercavitating vehicle must avoid an operational frequency of approximately 18 Hz, and the junction between the conical and cylindrical shells should also avoid frequencies between 15 Hz and 16 Hz.

**Keywords:** supercavitation; dynamic characteristics; structural response; nose-mounted cable towing system; tail-slap; FEM

## 1. Introduction

Underwater vehicles are significantly constrained in terms of maximum speed due to hydrodynamic frictional drag acting on their hulls. It is generally accepted that 40 m/s represents an ideal limit for such objects, with most practical applications failing to reach even half this value. Supercavitation technology liberates underwater vehicles from this speed constraint[1]. When an object travels through water at a sufficiently high velocity, the local pressure in the surrounding fluid drops below the vapor pressure, leading to the formation of vapor cavities. When this cavity flow nearly envelops the entire object, it is termed "supercavitation"[2]. Within a supercavity, the vast majority of the vehicle's surface does not contact the water directly, thereby reducing skin friction drag by approximately 90%. According to available reports, the Russian Navy's "Shkval" supercavitating torpedo achieves a velocity of 100 m/s[3]. Weapon systems utilizing supercavitation technology possess the potential to alter future naval warfare tactics; consequently, major

naval powers worldwide are actively engaged in research on supercavitation drag reduction.

Current research on supercavitation primarily focuses on the formation mechanisms, cavity length, and cavity generator shapes from a hydrodynamic perspective[4-7]. Some studies have begun to address the control of supercavitating vehicles and the oscillation of supercavities[8-9]. However, investigations into the structural characteristics of vehicles within a supercavitating flow remain scarce. Due to the unique operating environment, the stress state and structural dynamics of a supercavitating vehicle are entirely different from those of conventional underwater vehicles. In a supercavitating vehicle, the presence of hydrodynamic resistance, which is proportional to the square of the vehicle's velocity, makes the interaction between the water and the vehicle's nose cavitator particularly significant and potentially intense[10]. Consequently, the vehicle's hull will be subjected to high axial compressive loads. Furthermore, the transient and unstable nature of the cavity, its temporal variations, and its interaction with the vehicle generate lateral whipping impact loads. These factors necessitate a study of the structural behavior of supercavitating vehicles under time-varying forces. Therefore, analyzing and characterizing these interactions and their dynamic effects is essential for assessing the structural safety of supercavitating vehicles.

Previous research has explored various aspects of supercavitation. Some studies have focused on the characteristics of ventilated supercavities, examining flow patterns under different environmental conditions and the coupling effects between the jet and the cavity [11]. The methods for achieving stability control in ventilated supercavity flow are also presented. Numerical simulations have also been performed to understand the dynamic behavior of supercavitating vehicles, with researchers simulating different tail flapping frequencies and swinging amplitudes to analyze their hydrodynamic influence [12]. Furthermore, the design of visual control simulation experiments for supercavitating vehicle models has been explored to enhance understanding of control outcomes [13].

However, research on the structural analysis of supercavitating vehicles remains limited. Many studies employ simplified load modeling techniques and lack sufficient experimental validation. Addressing this gap is crucial for the safe and effective design of these vehicles. The study presented in this paper aims to address these issues by combining experimental and numerical analyses to provide a more comprehensive understanding of the structural response of supercavitating vehicles.

## **II. Experimental model of shell structure under supercavitating flow**

### **2.1 Model design**

The full-scale prototype of the model is made of high-strength titanium alloy. However, due to experimental constraints, a small-scale model is adopted. According to similarity criteria, the experimental results are

consistent under the same Froude number and cavitation number conditions. The experimental model is synthesized by combining a conical shell at the front with a cylindrical shell at the rear, and a disk-shaped cavitator is added to the front of the model to facilitate the generation of supercavitation during the experiment. The experimental model is manufactured in segments and assembled using threaded connections or adhesive bonding. Considering processing conditions and costs, the conical section of the model is machined from aluminum alloy, while the front cavitator is made of stainless steel. Based on trial calculations using empirical formulas for strain estimation[14], if metal were used as the material for the shell section, the measurement data from the strain gauges might be close to or below the sensitivity of the dynamic strain meter, making it difficult to obtain accurate readings. Therefore, the material for the shell section of the experimental model is determined to be a high-molecular synthetic material with a low elastic modulus. Additionally, considering the issue of attaching strain gauges, the final cylindrical shell section is made of polycarbonate material, i.e., PC material.

Due to its low elastic modulus, PC material is convenient for strain measurement, whereas aluminum, with its high elastic modulus, makes it difficult to measure strain values under low-speed conditions, and the model processing is also more challenging. However, both materials exhibit elastic behavior within the elastic range, and the dynamic response characteristics they reveal are the same. Considering the difficulty of bonding strain gauges to certain plastics, polycarbonate (PC) material was ultimately selected for the cylindrical shell section.

The cavitator is connected to the conical shell via threads, and the conical shell is connected to the connecting ring via threads as well. It is then bonded with adhesive to the cylindrical shell section, which is further bonded to the tail end cap. Since strain gauges need to be bonded to the inner surface of the cylindrical shell and the diameter is limited (inner diameter is only 61 mm), the entire cylindrical shell section is divided into three segments, each 90 mm in length, to facilitate the bonding process. After bonding the strain gauges to the inner surface of each segment, they are bonded together using adhesive and connecting rings to form the complete cylindrical shell section. A schematic diagram of the composition of the experimental model is shown in Figure 1.

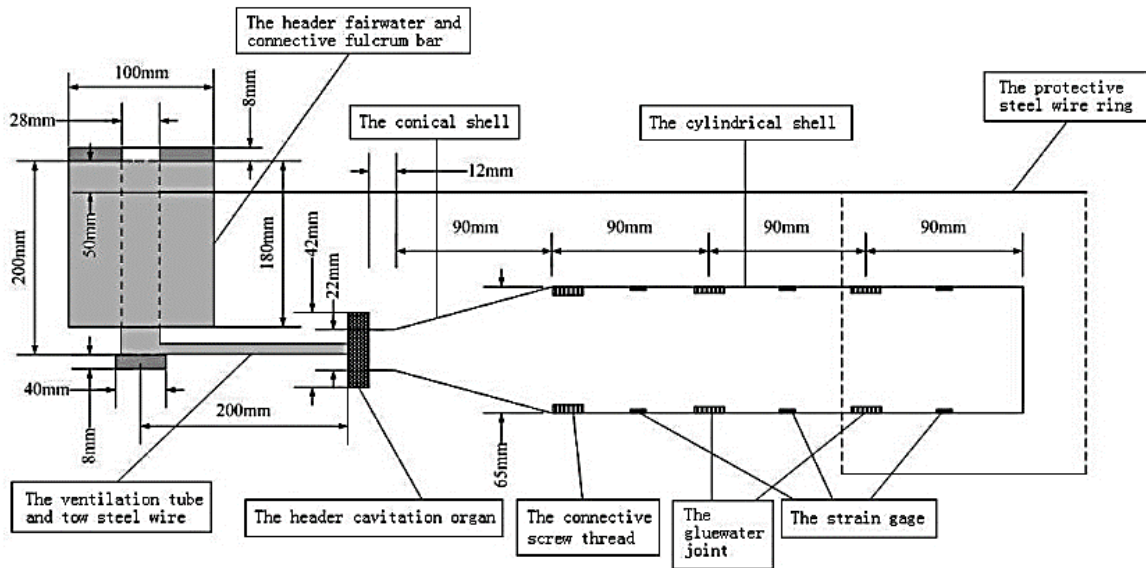


Figure 1 schematic diagram of the experimental model for the supercavitation vehicle

To measure the strain data of the shell structure during the experiment, strain gauges must be attached to the shell section. Since it is necessary to ensure that the external flow field of the shell's outer surface is not disturbed, the strain gauges must be attached to the inner surface of the shell. The DH3842 dynamic strain meter used for the test has 8 channels, which can simultaneously measure and record data from 8 channels. In combination with the actual conditions of this experiment the distribution of the strain gauge positions is shown in Figure 2. Strain gauges No. 1, 3, and 6 are located on the upper surface of the shell. Strain gauges No. 2, 4, and 7 are positioned on the side surface, while strain gauges No. 5 and 8 are situated on the lower surface.

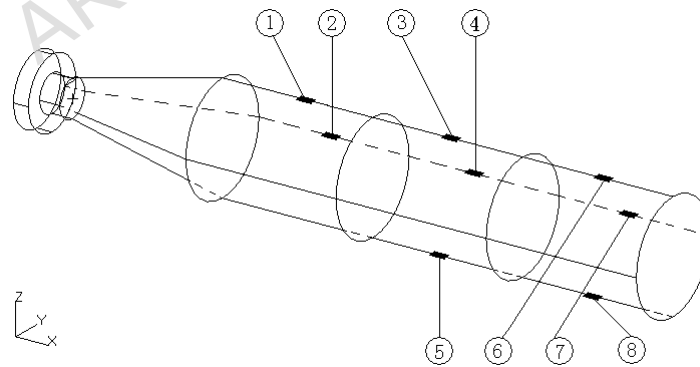


Figure 2 strain gauge position distribution

## 2.2 Experimental data and results

The experiment was conducted in the high-speed water tunnel of Northwestern Polytechnical University. The experimental model was connected to the water tunnel using a wire rope traction method at the front, and the cavitator at the front of the model was equipped with 16 ventilation holes with a diameter of 2.8 mm, evenly distributed circumferentially. The ventilation rate was maintained at a constant value

of  $0.01 \text{ m}^3/\text{s}$  to ensure stable supercavity formation. This rate was determined based on preliminary experiments to achieve a cavity that adequately envelops the vehicle body at the tested speeds. During the experiment, the supercavity phenomenon was clearly observed, and figures 3 and 4 show the comparison before and after the formation of the supercavity, respectively.

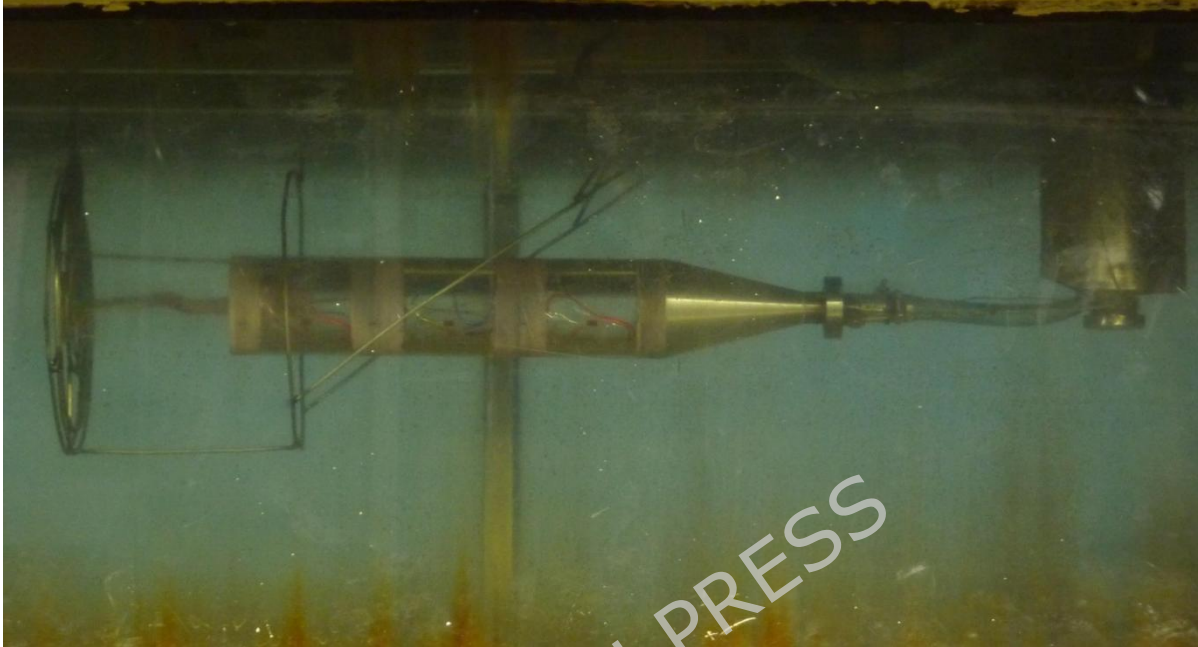


Figure 3 before the supercavity formed



Figure 4 after the supercavity formed

During the experiment, data from 8 channels were recorded. These data, which capture electrical signal characteristics, were converted into strain values through a specific formula, yielding the original strain values at each point. Since strain is induced by both axial force and tail-slapping vibration during supercavity formation, these two types of strain must be separated for individual analysis to understand their respective effects on the shell structure.



The following method was adopted: the two sets of original strain data obtained from symmetric directions were added together, and half of the result represents the strain value caused by axial force in that direction; conversely, the two sets of original strain data from symmetric directions were subtracted, and half of the result represents the strain value caused by vibration in that direction. From the perspective of strain gauge placement, gauges No. 3 and No. 5, as well as gauges No. 6 and No. 8, meet these requirements. Furthermore, to analyze the impact of velocity on the shell structure of the supercavitating vehicle, model experiments were conducted at three different velocities. By comparing the experimental results from these three velocities, some valuable patterns can be identified. For the sake of convenience in analysis, the comparative results for the three velocities, using strain gauges No. 3 and No. 5 as examples, are shown in Figures 5 and 6.

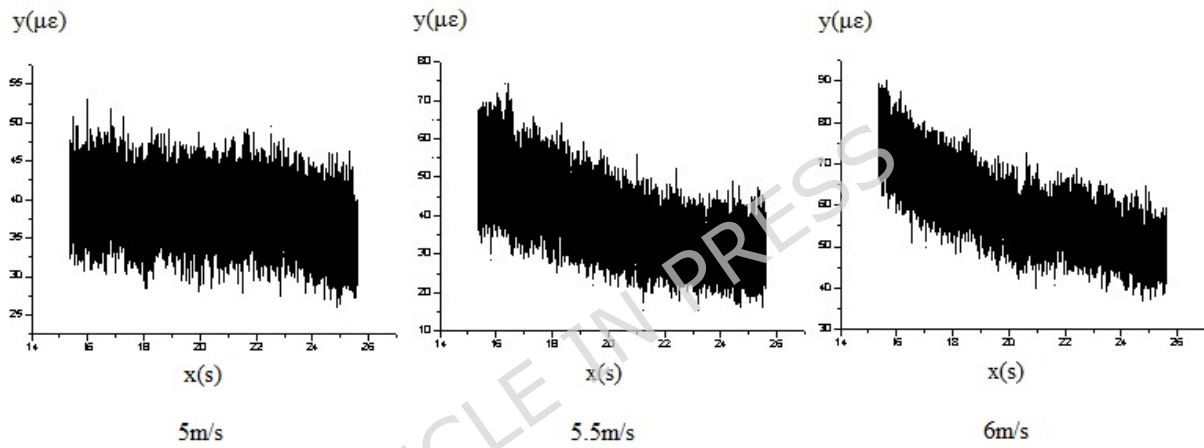


Figure 5 comparison of axial results at three different speeds

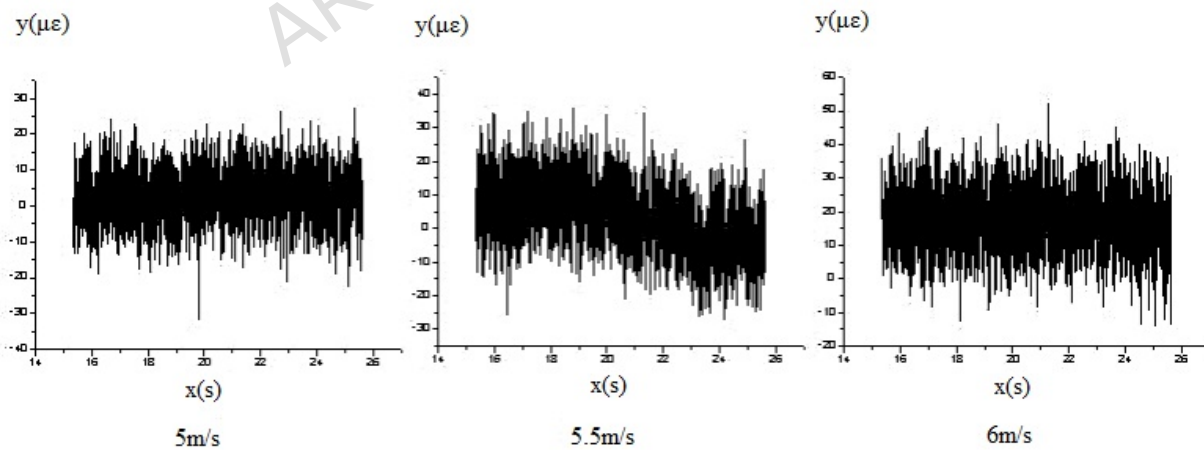


Figure 6 comparison of vibration results at three different speeds

As illustrated in Figures 5 and 6, the axial strain exhibits a decaying trend, whereas the strain induced by tail-slapping vibration demonstrates periodic characteristics with a relatively stable amplitude. Higher velocities result in more severe attenuation of axial strain, indicating a more pronounced drag reduction effect. Conversely, the amplitude of

strain caused by tail-slapping vibration increases with velocity, suggesting that the structural impact of tail-slapping becomes more significant at higher speeds. Specifically, at a velocity of 6 m/s, the strain amplitude induced by tail-slapping is nearly comparable to that induced by axial force, implying that both factors jointly constitute the primary influences on the structure. Therefore, the effect of tail-slapping on the shell structure of supercavitating vehicles at even higher velocities warrants further in-depth investigation. However, due to limitations in current experimental capabilities, high-velocity supercavity experiments are not yet feasible. Consequently, the Finite Element Method (FEM) must be employed to analyze the structural response under such high-speed conditions.

### III. Finite element description

#### 3.1 Finite element governing equations

Based on Hamilton's principle, the finite element governing equations for the supercavitating underwater vehicle can be expressed as

$$[M]\{\ddot{u}\} + [C]\{\dot{u}\} + [K]\{u\} = \{F_d(t)\} \quad (1)$$

where,  $[M]$  represents the structural mass matrix,  $[C]$  represents the structural damping matrix,  $[K]$  represents the structural stiffness matrix,  $\{u\}$  represents the displacement vector and  $\{F_d(t)\}$  represents the external load vector.

The various components of the system can be assembled from their respective element matrices, i.e.,

$$[M] = \sum [M^{(e)}] \quad [C] = \sum [C^{(e)}] \quad [K] = \sum [K^{(e)}] \quad (2)$$

Neglecting the effects of damping and external loads, the governing equation can be simplified to

$$[M]\{\ddot{u}(t)\} + [K]\{u(t)\} = 0 \quad (3)$$

Equation (3) is the free vibration equation of the system.

#### 3.2 Finite element model

The Finite Element Model consists of the following elements: shell elements are used to simulate the outer shell of the supercavitating vehicle, and the analysis is conducted using the general-purpose commercial finite element software. The software is MSC.NASTRAN 2020. The URL link is <https://www.gopaysoft.com/?p=14936>. The main dimensions are as follows: total length of 3.5 m, with a cylindrical shell section of 2.8 m and a conical shell nose section of 0.7 m; the cross-sectional diameter of the cylindrical shell is 0.21 m, and the nose cross-sectional diameter of the conical shell is 0.07 m; the shell thickness



is 0.006 m. The finite element model established accordingly is shown in Figure 7, containing a total of 3682 nodes and 3680 elements. The material used for this model is aluminum.

We specify the use of quadrilateral shell elements (SHELL elements in NASTRAN) and the mapped meshing technique. We tested three mesh densities and compared the maximum stress at a critical location. The results confirmed that the chosen medium-density mesh (3680 elements) yields results within 5% of the finer mesh, ensuring mesh independence.

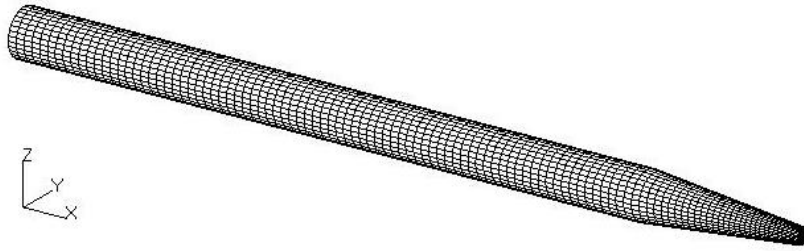


Figure 7 a finite element model of the supercavitating underwater vehicle

### 3.3 Determination of external loads

During the navigation of a supercavitating underwater vehicle, it is subjected to external loads. These external loads mainly consist of three forces: propeller thrust  $F_p$ , fluid resistance  $F_{FR}$ , and tail-slapping impact force  $F_I$ . The distribution and application of these external loads are illustrated in Figure 8.

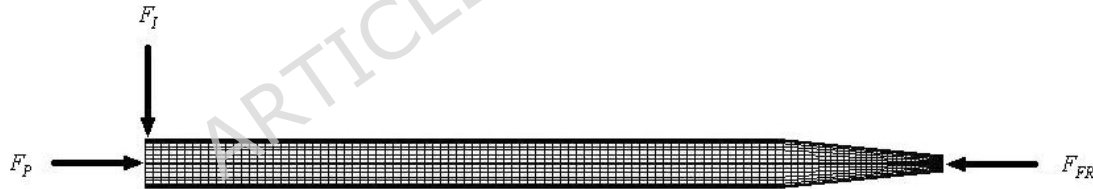


Figure 8 schematic diagram of forces on the supercavitating vehicle

#### 3.3.1 Axial dynamic load

The model is based on the following assumptions. It is assumed that the cavity is approximately symmetric about the horizontal line. It is assumed that the trajectory of the vehicle's center of mass is approximately a straight line. It is assumed that the rotation of the supercavitating vehicle about its axis of symmetry is negligible during operation[14]. It is assumed that the propeller thrust is approximately constant.

The magnitude of the fluid resistance can be expressed as

$$F_{FR} = \frac{1}{2} \rho A v^2 k \cos^2 \theta \quad [4]$$

Where,  $\rho$  is the water density,  $A$  is the frontal cross-sectional area,  $v$  is the forward speed of the vehicle,  $k$  is a dimensionless constant,  $\theta$  is the angle between the vehicle's axis and the cavity's axis[15].

According to Newton's second law, the equation of motion can be derived.

$$m\ddot{x} = F_p - F_{FR} \quad [5]$$

Substituting equation (4) into equation (5), we obtain

$$m\ddot{x} = F_p - \frac{1}{2}\rho A \dot{x}^2 k \cos^2 \theta \quad [6]$$

Where  $x$  is the displacement of the vehicle's center of mass along the forward direction.

For small angles, equation (6) can be written as

$$\ddot{x} = \frac{F_p}{m} - \frac{\rho A k}{2m} \dot{x}^2 \quad [7]$$

We can get equation (8) if we set  $\xi = \frac{\rho A k}{2m}$

$$\ddot{x} + \xi \dot{x}^2 = \frac{F_p}{m} \quad [8]$$

Equation (8) has the solution as follows.

$$\dot{x}(t) = \beta \frac{e^{2\alpha t} - 1}{e^{2\alpha t} + 1} \quad [9]$$

Where  $\alpha = \sqrt{\frac{F_p \xi}{m}}$ ,  $\beta = \sqrt{\frac{F_p}{m \xi}}$ .

So the expression of the flow resistance is as follows.

$$F_{FR}(t) = \frac{1}{2} \rho A k \left( \frac{e^{2\alpha t} - 1}{e^{2\alpha t} + 1} \beta \right)^2 \quad [10]$$

The following parameter values are used in the calculation:

$$r = 1 \times 10^3 \text{ kg/m}^3, \quad A = \pi (0.035)^2 \text{ m}^2, \quad k = 0.9, \quad m = 164 \text{ kg}, \quad F_p = 1.732 \times 10^4 \text{ N}.$$

The fluid resistance curve determined accordingly is shown in Figure 9.

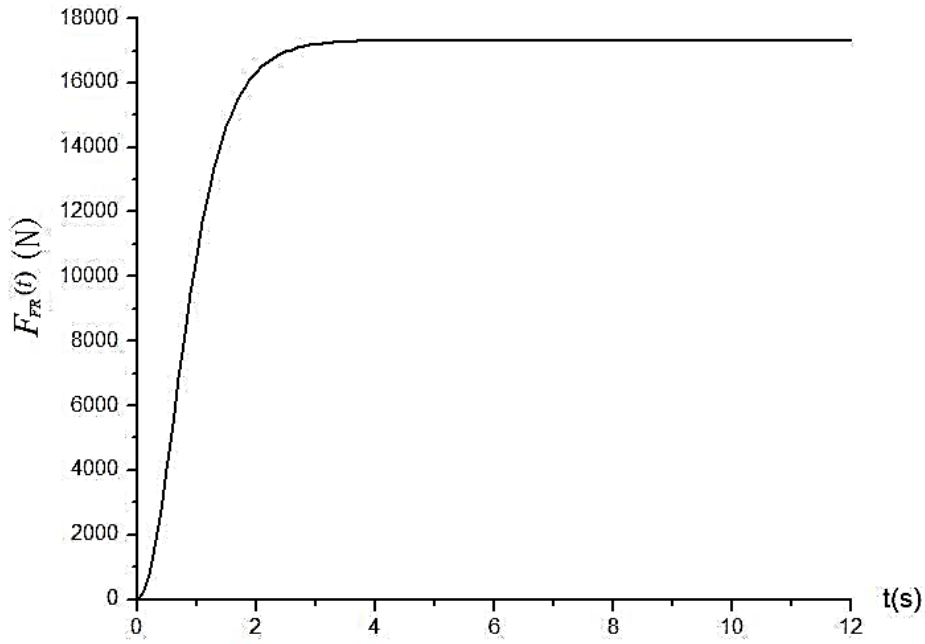


Figure 9 Fluid resistance characteristic curve

### 3.3.2 Tail-induced impact load

A simplified model is proposed to determine the tail-induced impact load, where the magnitude of the impact force is assumed to be proportional to the distance  $\Delta$  by which the tail is immersed in the fluid, and this force is applied at the tail section of the vehicle, as illustrated in Figure 10.

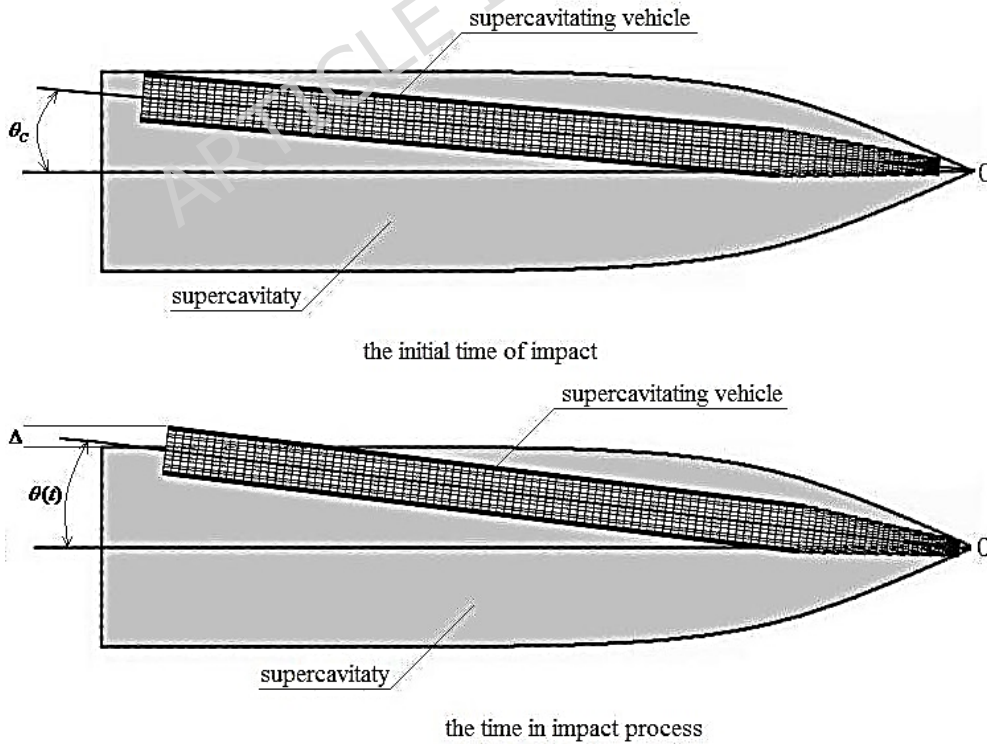


Figure 10 a simplified model of the impact process

According to the aforementioned assumptions, the impact force can be described as follows.

$$F_i(t) = K(\theta) \quad (11)$$

The function referenced in Equation (11) is defined as follows.

$$K(\theta) = \begin{cases} K_0 & |\theta(t)| \geq \theta_C \\ 0 & |\theta(t)| < \theta_C \end{cases} \quad (12)$$

In Equation (12), the critical angle  $\theta_C$  represents the initial impact condition between the vehicle and the vapor cavity, while the constant  $K_0$  defines the duration of the impact.

The duration of a single impact is assumed to be  $T$ . The relationship between  $T$  and  $K_0$  can be derived by considering the rigid body motion of the supercavitating vehicle. For small-amplitude vibrations in the plane, the vehicle's motion equation can be expressed as follows.

$$I_z \ddot{\theta}(t) + K(\theta) L^2 \theta(t) = K(\theta) L^2 \theta_C \quad (13)$$

In Equation (13),  $I_z$  is the moment of inertia, and  $L$  is the vehicle length,  $K(\theta)$  is as described in Equation (12). For the short-duration impact event of tail-slap (on the order of milliseconds), the effect of structural damping is negligible compared to the inertial and stiffness forces.

Equation (13) has the solution as follows.

$$\theta(t) = \theta_C + \frac{\dot{\theta}_0}{\omega} \sin(\omega t) \quad (14)$$

In equation (14)  $\dot{\theta}_0$  is the initial angular velocity of the vehicle and the range of  $t$  is from zero to  $T$ ,  $\omega = \sqrt{\frac{K_0}{I_z}} L$ .

Imposing  $q(T) = q_c$  to equation (14) we can get

$$T = \frac{\pi}{\omega} = \frac{\pi}{L} \sqrt{\frac{I_z}{K_0}} \quad (15)$$

Similarly we can get the maximal penetration distance  $D_m$ .

$$\Delta_m = \frac{\dot{\theta}_0}{\omega} = \frac{\dot{\theta}_0}{L} \sqrt{\frac{I_z}{K_0}} \quad (16)$$

Combining the results of previous supercavitation experiments, the final tail-slap impact load characteristic curve is presented in Figure 11.

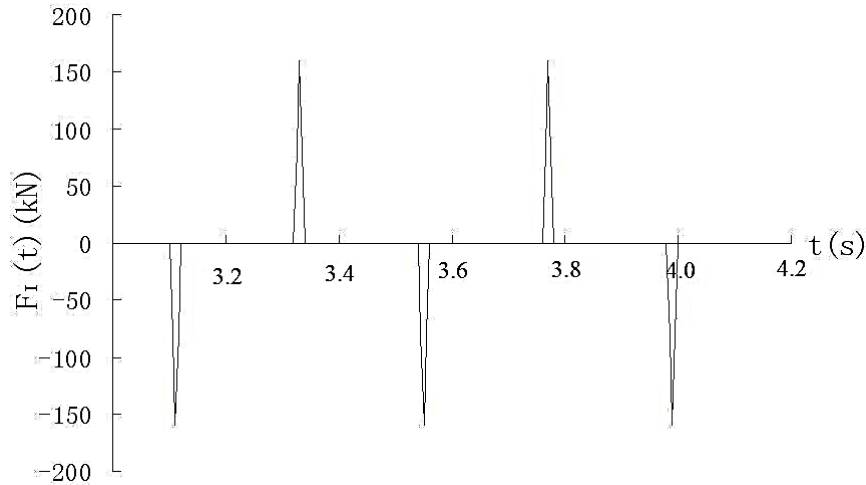


Figure 11 tail-slap impact load characteristic curve

It is important to acknowledge the limitations of this simplified impact model. The model assumes a rigid body and a proportional relationship between impact force and immersion depth, neglecting the complexities of fluid-structure interaction. In reality, the flow field around the tail in supercavities is highly transient and asymmetric, with potential for drastic changes in fluid dynamic pressure and local bubble collapse during tail-slapping. This can lead to higher frequency and amplitude impact loads than predicted by the model. The applicability of this model is best suited for preliminary analysis and qualitative understanding. For higher speeds or more complex conditions, a more sophisticated approach, such as bidirectional fluid-structure interaction (FSI) modeling, would be necessary for future work.

#### IV. Numerical results and discussion

##### 4.1 Comparison of experimental and numerical results

To further analyze the impact of shock loading on the structure, numerical calculations were performed on the experimental model using the Modal Superposition Method. The calculated results were processed by taking half the difference between the two strain values at symmetric locations, yielding the strain induced by the shock load at that position. The results obtained via the modal superposition method are of the same order of magnitude as the experimental results but are slightly higher, exhibiting several distinct peaks. To specifically analyze the differences and correlations between the two, the numerical method was compared with the experimental results, with a detailed comparison provided in Table 1.

Table 1 The comparison of the numerical results and experimental results

|                      | The maximal strain of stain gages numbered 3 and 5 which is caused by tail impact | The maximal strain of stain gages numbered 6 and 8 which is caused by tail impact |
|----------------------|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------|
| Experimental results | 52.22                                                                             | 30.21                                                                             |

| $\mu\epsilon$                           |        |       |
|-----------------------------------------|--------|-------|
| Numerical results $\mu\epsilon$         | 68.80  | 30.98 |
| Difference of two results $\mu\epsilon$ | 16.58  | 0.77  |
| Percentage                              | 31.75% | 2.55% |

Based on Table 1, it can be observed that the numerical results are of the same order of magnitude as the experimental data. When using the modal superposition method, the maximum strain caused by the impact load shows a difference of 2.55% between the calculated and experimental results at the locations of strain gauges No. 6 and No. 8. However, at the locations of strain gauges No. 3 and No. 5, the discrepancy reaches 31.75%. Given that the experimental conditions cannot be perfectly replicated in the numerical simulation, such deviations are considered acceptable. Specifically, factors such as slight variations in material properties between the physical model and the numerical representation, as well as minor imperfections in the experimental setup (e.g., cable alignment), can contribute to these discrepancies. The modal superposition method relies on accurate modal shapes and frequencies, which, while validated by the overall agreement, can still exhibit local variations. Furthermore, the fact that the supercavity did not completely envelop the entire tail section during the experiment may also contribute to these differences. Overall, the results obtained via the modal superposition method are higher than the experimental values. From a structural design perspective, this overestimation indicates that the method is conservative and thus safer. Therefore, the calculated loads and numerical methods adopted in this study are deemed feasible.

## 4.2 Numerical results

To provide a more comprehensive understanding, Figure 12 shows the overall stress distribution on the vehicle at a specific time point ( $t=3.5s$ ) during the tail-slap event, confirming the high stress concentration in the cylindrical shell mid-section.



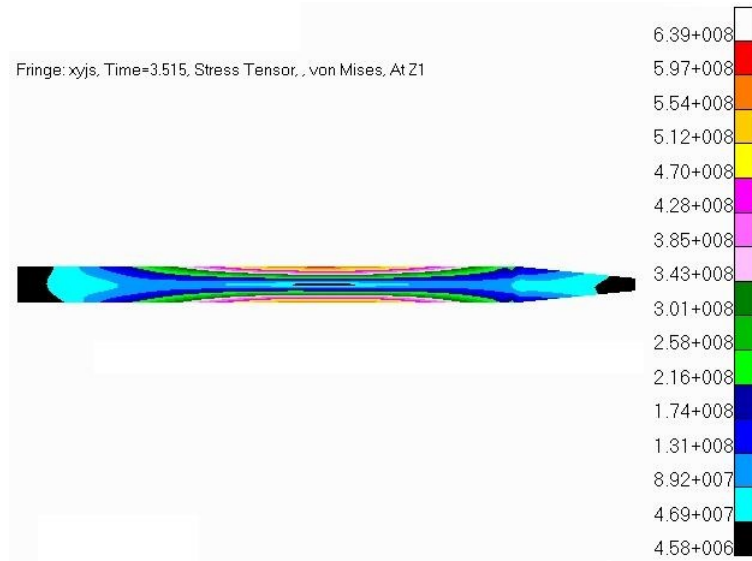


Figure 12 Overall von Mises stress distribution on the vehicle structure at  $t=3.5s$

The dynamic response problem of supercavitating underwater vehicles under combined dynamic axial and tail-slap impact loads is investigated, and the obtained results provide valuable reference for the structural design of supercavitating vehicles. Figures 13 to 15 present the stress variation curves for various typical sections of the vehicle.

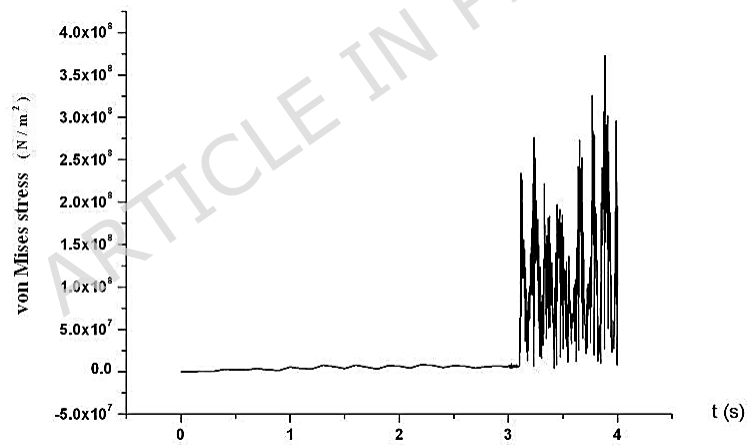


Figure 13 conical shell mid-section stress variation curve

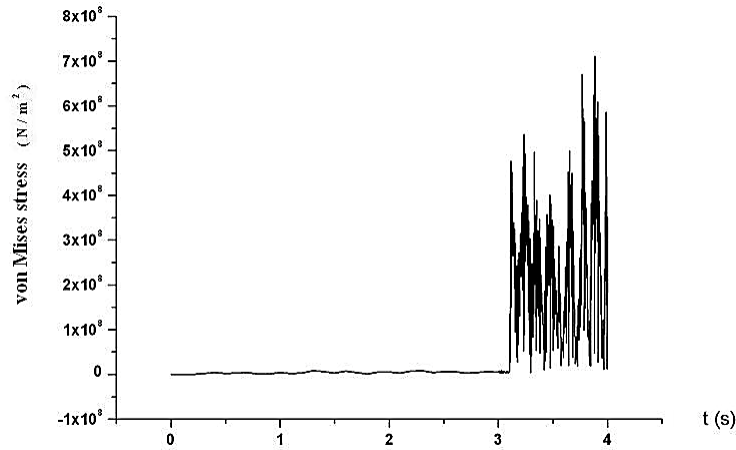


Figure 14 conical-cylindrical shell junction stress variation curve

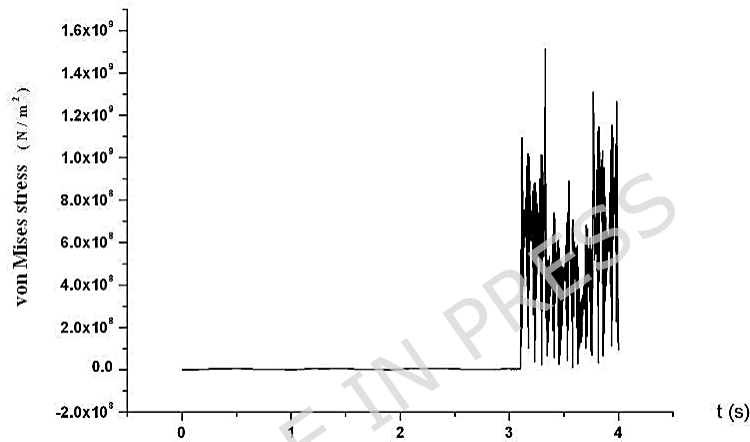


Figure 15 cylindrical shell mid-section stress variation curve

As can be seen from Figures 13 to 15, the stress generated by the supercavitating underwater vehicle during the tail-slap phase is significantly greater than that during the initial acceleration phase. During the tail-slap process, the stress in the cylindrical shell section of the vehicle reaches its maximum value, with the stress amplitude in the mid-section of the cylindrical shell exceeding 1000 MPa. This value far surpasses the material's strength limit, inevitably leading to structural failure in the cylindrical shell section. The stress results presented are the von Mises stress amplitudes at specific points (primarily the mid-section of the cylindrical shell). The high stress concentration in the cylindrical shell during tail-slap is primarily due to its location relative to the impact force. The tail-slap, acting as a concentrated load, induces bending moments and shear forces in the shell, with the maximum bending stress occurring at the mid-section. This is further exacerbated by the relatively thin wall thickness of the shell, making it susceptible to buckling and high stress concentrations.

In addition to analyzing the stress distribution results, we also examined the acceleration variations resulting from the dynamic response of the supercavitating underwater vehicle. Figures 16 to 18 present the

time-dependent acceleration curves for various sections of the vehicle. The acceleration discussed is the lateral acceleration at the vehicle's center of mass.

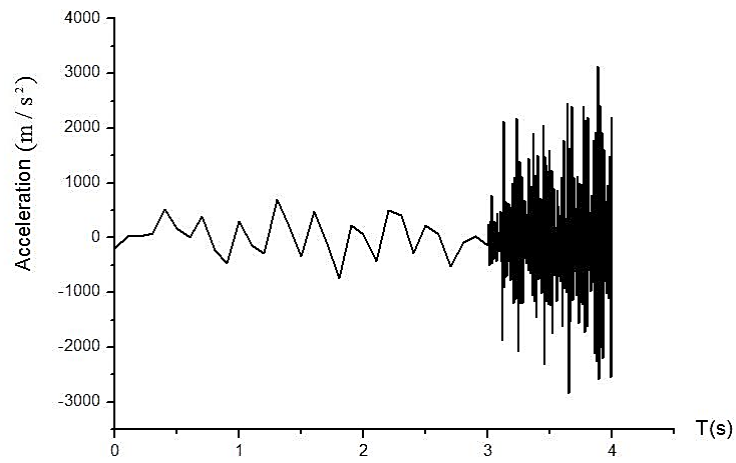


Figure 16 time-dependent acceleration curve for the mid-section of the conical shell

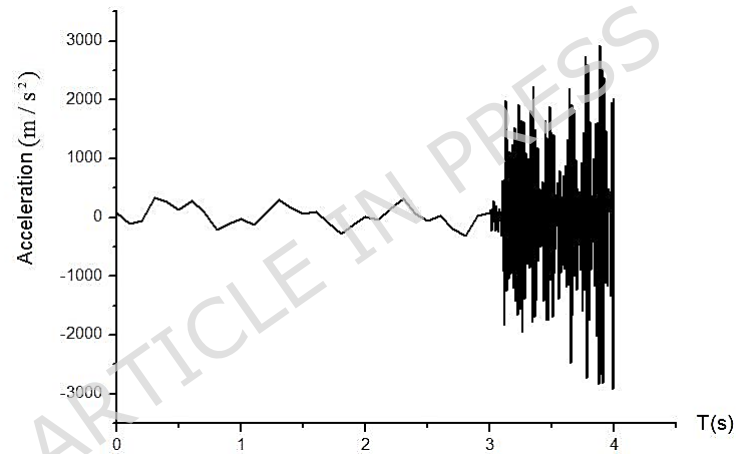


Figure 17 time-dependent acceleration curve for the junction of conical and cylindrical shells

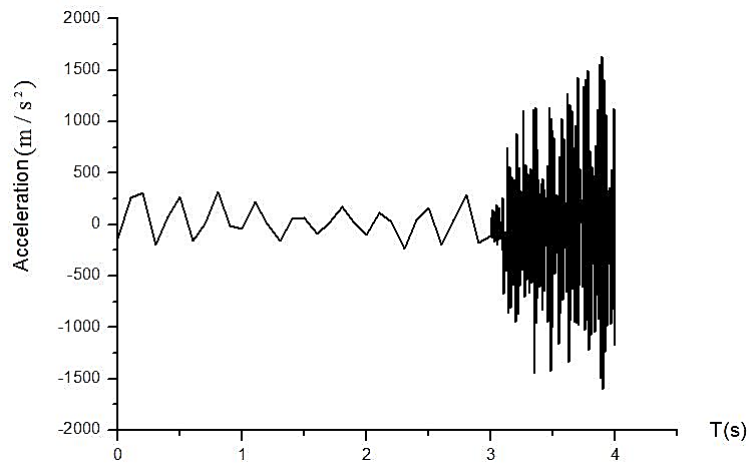


Figure 18 time-dependent acceleration curve for the mid-section of the cylindrical shell

Figures 16 to 18 indicate that the acceleration during the tail stage is significantly larger than that during the initial stage, with the acceleration amplitude showing an accumulative trend. The supercavitating underwater vehicle experiences substantial overload during the tail stage, reaching a maximum magnitude of 300g. The Fourier transform of these results is presented in Figures 19 to 21.

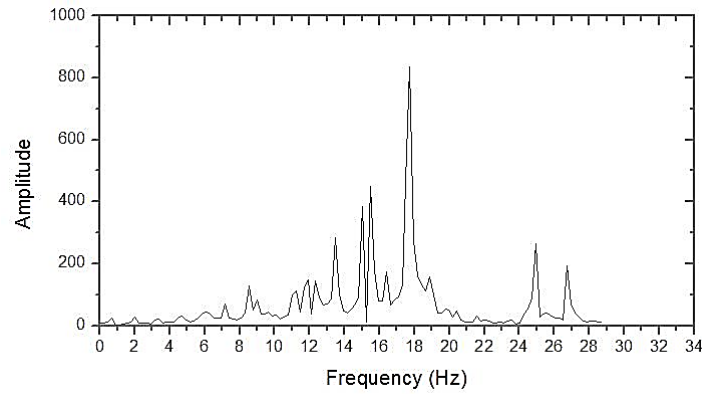


Figure 19 Fourier transform results(4 seconds) in mid-section of the conical shell

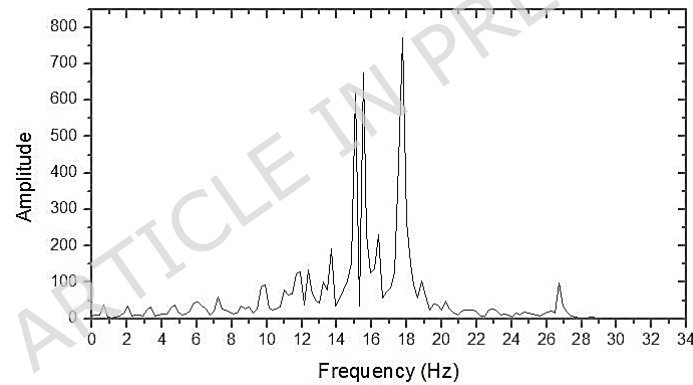


Figure 20 Fourier transform results(4 seconds) in junction of conical and cylindrical shells

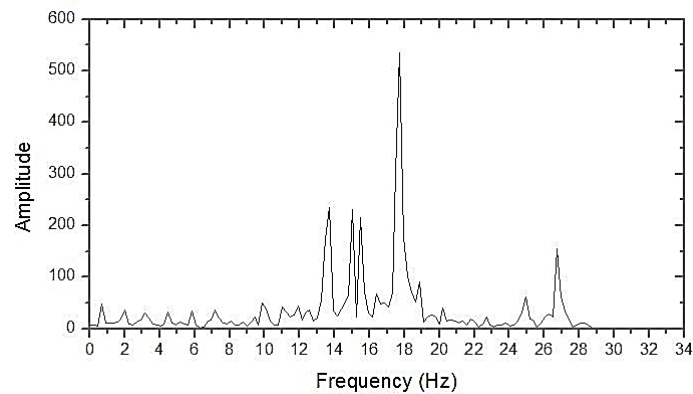


Figure 21 Fourier transform results(4 seconds) in mid-section of the cylindrical shell

To understand the source of these characteristic frequencies, a modal analysis of the vehicle structure was performed. The first few natural frequencies were found to be: 12.8 Hz, 15.1 Hz, 18.3 Hz, 24.9 Hz, and 27.5 Hz. Comparing these with the peaks in the FFT results reveals a strong correlation. The peaks at approximately 15 Hz, 18 Hz, and 27 Hz correspond closely to the structural natural frequencies, indicating that the tail-slap excitation is inducing resonance. This resonance significantly amplifies the dynamic response at these frequencies, justifying the recommendation to avoid operational speeds that excite these modes.

The material differences between the model and the prototype will affect the predicted natural frequencies to some extent. However, the patterns are consistent within the elastic range. The scaled model is adopted because its size is suitable for experiments in a high-speed water tunnel, but it cannot replace the results of a full-scale model. This is also an area we need to work on in the future.

## **V. Conclusion**

The experimental investigation on the structural model of a supercavitating vehicle was conducted for the first time using a header wire rope towing method at three different speeds. The results demonstrate that the supercavitating experiment using the header wire rope towing method is feasible. The higher the speed, the more significant the drag reduction effect of the supercavitating vehicle; however, the strain amplitude caused by tail-slapping vibration increases with speed, making the impact of tail-slapping on the structure more pronounced. Based on the supercavitating experiment, the characteristics of dynamic loads acting on the supercavitating vehicle were described, and the dynamic response characteristics of the underwater supercavitating vehicle under combined loads were studied using the finite element method.

The numerical results reveal that the stress generated during the tail-slapping stage is significantly higher than that during the initial acceleration stage. The stress in the cylindrical shell section reaches its maximum, with an amplitude exceeding 1000 MPa at the midsection, which far surpasses the material's strength limit and will lead to structural failure of the cylindrical shell. The acceleration during the tail-slapping stage is much higher than in the initial stage, and the acceleration amplitude shows a cumulative trend, subjecting the vehicle to a large overload of up to 300g.

Further results after Fourier transform indicate that all parts of the underwater supercavitating vehicle must avoid the frequency around 18 Hz, and the junction of the conical and cylindrical shells should also avoid the working frequency range of 15 Hz to 16 Hz. From a structural safety perspective, the structure of the midsection of the cylindrical shell requires further reinforcement.

## References

- [1] Ma Yiming, Tang Kui, Kong Lingquan, et al. Response of underwater cylindrical shell subjected to supercavitating projectile trans-medium penetration: Experiment and simulation. *Defence Technology*, 2025, 50(8): 304-323.
- [2] Cui Naigang, Chen Liang, Cao Jiamu, et al. Review of Drag-Reduction Techniques for Underwater Vehicles. *Astronautical Systems Engineering Technology*, 2023, 7(1): 1-13.
- [3] Wang Kangjian, Xia Youran, Kang Qizhuang, and Wu Youli. Supercavitating Projectile Tail-Slaps Based on a Normal Distribution. *Journal of Marine Science and Application*, 2025, 24(2): 398-414.
- [4] Tian Jianhui, Hu Chenming. Numerical simulation of high speed projectile discharge process. *Journal of Ordnance Equipment Engineering*, 2023, 44(12): 79-84, 182.
- [5] Li Peng, Chen Weiye, Wang Zhi, et al. Experimental study of supercavity stability and ventilation rate in high speed water tunnel. *Journal of Ship Mechanics*, 2024, 28(4): 3-8.
- [6] Peng Ruizhe, Feng Reying, Xiang Min, et al. A study on the vertical water entry performance of a head jet supercavitating navigation body. *Journal of Vibration and Shock*, 2024, 43(20): 245-253.
- [7] Wei Ping, Weng Mingdeng, Yang Xiaobin, et al. Research on the Supercavitation Characteristics of a High-speed Vehicle Near the Free Surface. *Acta Armamentarii*, 2025, 46(5): 100-111.
- [8] Sun Tiezhi, Wang Shisheng, Xie Bohan, et al. Review of Numerical Calculation Research on Underwater Supercavitation Flow. *Journal of Unmanned Undersea Systems*, 2023, 31(3): 349-364.
- [9] Lv Keyu, Zhang Huanhao, Jin Xiaoyu, et al. Influence of Transverse Flow on a Supercavitating Projectile with Tail Fin during Water Entry. *Acta Armamentarii*, 2025, 46(5): 419-430.
- [10] Qi Xiaobin, Yao Zhong, Li Ruijie, et al. Application of Supercavitation Technology in Anti-submarine Warfare. *Digital Ocean & Underwater Warfare*, 2022, 5(2): 109-114.
- [11] Wang Zhiying, Wang Jingzhu, Huang Jian, et al. Research progress on the stability mechanism and control of ventilated supercavitation. *Advances in Mechanics*, 2025, 55(1): 175-217.
- [12] Zhang Ke, Cai Xiaowei, Wang Zhi, et al. Numerical Study on Hydrodynamic Influence of Tail-slap Parameters of Supercavitating Vehicles. *Digital Ocean & Underwater Warfare*, 2025, 8(3): 301-311.
- [13] Bai Tao, Wang Xingjie, Han Yuntao. Design of visual control simulation experiment for supercavitating vehicle. *Laboratory Science*, 2024, 27(1): 66-71.
- [14] Huang Chuang, Guo Kaixin, Qin Kan, Luo Kai, Li Daijin, Dang Jianjun. Hydrodynamic Characteristics and Supercavity Shape of Supercavitating Projectiles Launched with Supersonic Speed. *Journal of Marine Science and Application*, 2022, 21(2): 24-33.
- [15] Yu Yilian, Shi Yao, Pan Guang, Che Peiqi. Numerical analysis of cavitation

and impact load characteristics of supercavitating vehicle entering water at high speed. Journal of Northwestern Polytechnical University, 2022, 40(3): 584-591.

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